AI AND DESIGN — FINAL POSTER

SG Guard

A trust-first defence against AI-personalised scams — built for Singapore's seniors, not around them.

Carwyn Yeo · Lam Yan Yee
AI and Design
Final Poster — 2026

01 Abstract

Singapore's seniors are digitally comfortable but not digitally confident, and generative AI has widened that gap into a public-safety crisis: government-impersonation scams rose 124% year-on-year, and elderly victims lose an average of S$37,053 each, manipulated into transferring money themselves rather than being hacked. Static defences such as ScamShield's blacklists cannot keep pace with an adversary that now drafts a personalised script in seconds and clones a voice from three seconds of audio.

SG Guard reframes the contest: instead of racing generative AI on detection speed, it uses AI to support judgement. Seniors share a suspicious message by typing, photographing, or speaking it, and a schema-locked Gemini backend reads it for manipulation tactics rather than keywords, returning one of three honest verdicts — warning, unsure, or safe — each explained in plain language and always ending in a one-tap handoff to a trusted person. The interface borrows the vocabulary of Singapore's physical priority infrastructure, treating digital safety as civic infrastructure rather than a personal gadget a senior must master alone. The project's central argument: trustworthy AI safety tools must be honest about their own limits, not just accurate about the threats they detect.

Fig. 1 — the design lens

AN AI ARMS RACE, REFRAMED

THEIR AI
scaling deception, cheaply, at scale

- Harvests leaked data & social media to profile one target
- Drafts a personalised script in seconds, no expertise needed
- Clones a convincing voice from roughly three seconds of audio
- Rents as "scam-as-a-service" — kits from about US$60 / month

Wins by being faster than blacklists

NOT A SPEED RACE

OUR AI
supporting a senior's own judgement

- Reads for manipulation tactics, not a keyword blacklist
- Multimodal — checks typed text, photos, and speech alike
- Answers warning / unsure / safe — never fakes certainty
- Explains each tactic in plain language, then hands off to a person

Wins by staying honest under pressure

How to read this: generative AI now equips scammers to personalise deception at a pace static blacklists can't match. SG Guard doesn't try to out-run that speed — it changes what the contest is measured on, from detection speed to legible, honest judgement support.

02 Design Process

Research began with a stakeholder map spanning scam syndicates, government agencies, banks and telcos, family members, and People's Association grassroots networks. One voice was largely absent from how existing anti-scam tools get designed: seniors' own. This gap, paired with evidence that blacklists can't rewrite themselves as fast as an attacker's AI rewrites a message, set the brief — build judgement support, not just detection.

Early iterations tested a hard "reject" screen that blocked suspicious messages outright. Feedback and participatory-design literature (Zhai et al., 2025) showed seniors disengage the moment a tool feels like it's taking judgement away rather than supporting it, so the block was dropped for a three-way verdict that explains its reasoning instead of issuing a command. "Unsure" was added as an AI-fairness decision: forcing a binary verdict onto an ambiguous message manufactures false confidence the senior would pay for.

The final intervention is a multimodal web portal: type, photograph, or speak a message, never forcing one input on someone least comfortable with it. It's sent to an Express server, forwarded to the Gemini API with a hard-constrained JSON schema — so the verdict is a true enum, not free text a model could drift on — and returned as a plain-language, tactic-by-tactic explanation with a permanent "speak to a person" exit. A mid-project honesty audit also caught a false privacy claim ("never leaves this screen") and rewrote it to describe what actually happens to the data, and de-gendered the trusted-contact default from a fixed "(Daughter)" label.

Fig. 2 — how the intervention works

01
Senior shares a message
— typed, photographed, or spoken aloud

02
Express server (server.js)
— holds the API key; forwards only this one item

03
Gemini API + system prompt
— reads for tactics, not keywords

04
Structured JSON — verdict is an enum
— warning / unsure / safe, schema-locked

05
Plain-language result, tactic by tactic
— with a one-tap human exit

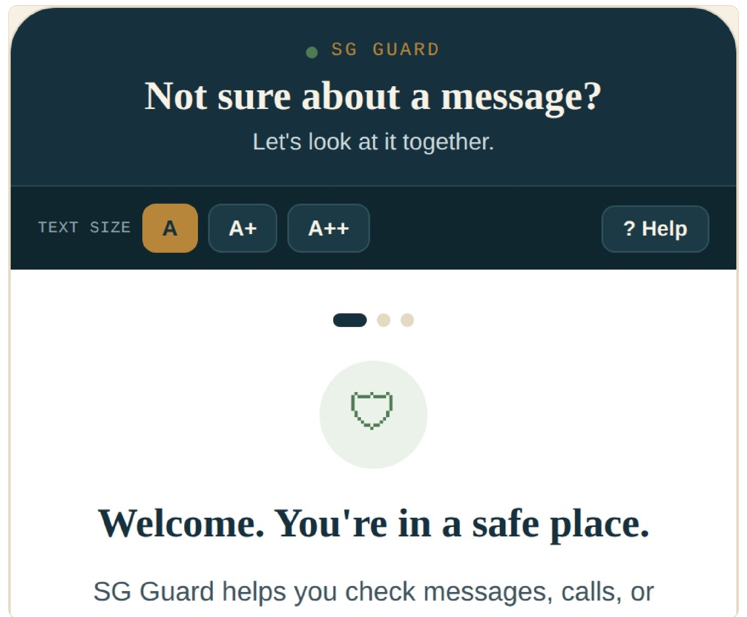
Nothing is stored server-side; nothing is logged with identity attached.
The front-end honesty fix describes a pipeline that is actually true on the back end.

How to read this: a hard-constrained response schema is what makes a calibrated third verdict — "unsure" — safe to ship, rather than a state the model could silently collapse out of.

03 How It Works

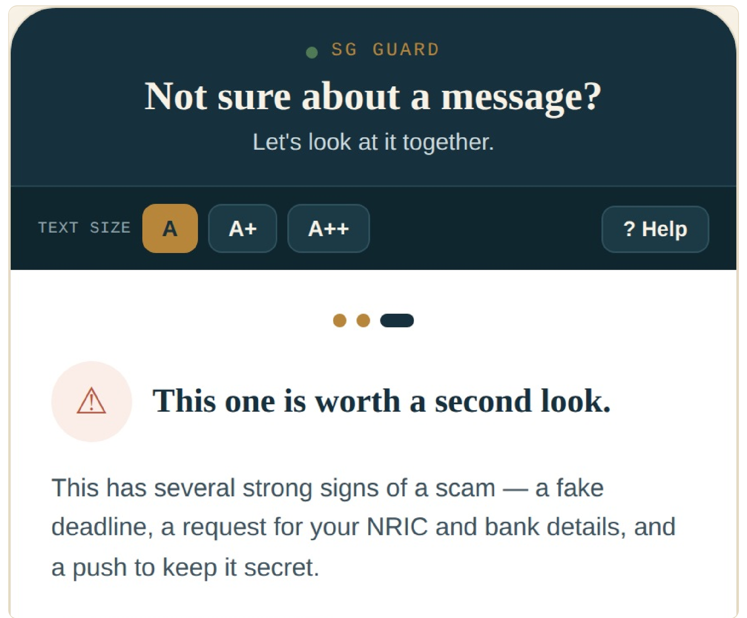
Walkthrough of the live prototype

SG Guard is a working multimodal prototype wired to a live Gemini backend, not a mockup. The four screens below show the actual flow a senior moves through, from opening the portal to receiving an honest, explained verdict.

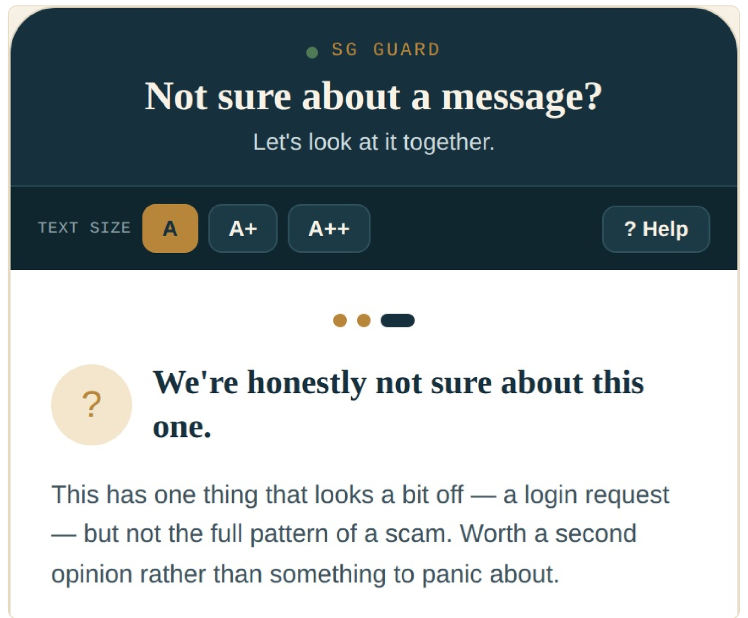


Screen 1 — Welcome
States the promise upfront
Names what the portal does and doesn't do before asking for anything — trust is established first.

Screen 2 — Share
Three ways in, one flow
Type, upload a photo, or speak — never forced into the one input method trusted least.



Screen 3 — Warning
Explains, doesn't just alarm
Each flagged tactic is named in plain language, with the exact phrase highlighted.



Screen 4 — Unsure
The honest third answer
Lists what's concerning and what's reassuring side by side, not false confidence.

04 Where This Sits in the Studio

SG Guard against the semester's shared vocabulary

Participatory Design
The three-way verdict exists because an early hard "reject" screen was dropped after feedback and participatory-design literature (Zhai et al., 2025) showed it disengaged seniors by taking judgement away rather than supporting it. The project stays honest about the limits of that participation too: no formal testing session with seniors at an Active Ageing Centre has taken place yet.

AI Fairness
"Unsure" is a fairness decision, not a UX nicety. Forcing a binary warning/safe verdict onto a genuinely ambiguous message would manufacture false confidence the senior — not the system — would pay for. The schema-locked response keeps that third state a real option the model can reach.

Surveillance, Privacy & AI
A mid-project honesty audit caught the prototype's own privacy claim — that a message "never leaves this screen" — describing something the backend didn't actually do. The copy was rewritten to say what really happens: sent to an AI model, checked, then deleted.

Adversarial Design
The project is framed explicitly around an adversary: an asymmetric, ongoing arms race between AI-personalised scam generation and scam detection (Fig. 1). SG Guard's response changes what the contest is measured on, from detection speed to legible, honest judgement support.

05 Discussion & Limitations

SG Guard's strengths are its honesty: interface copy that matches what the backend actually does, calibrated uncertainty instead of false binary confidence, true multimodal accessibility, and a human handoff on every result. Its central limitation is honesty-shaped too: no formal participatory testing with seniors has yet taken place. The tool is English-only, its "add someone you trust" feature is a placeholder, and cost/latency at scale remain unvalidated. The AI arms race it addresses is asymmetric and ongoing — SG Guard is a living system, not a finished artefact.

Fig. 3 — an honest scorecard

Strengths
- ✓ Interface copy matches backend reality
- ✓ Calibrated uncertainty, not false confidence
- ✓ True multimodal access — type, photo, voice
- ✓ Human handoff on every result

Limitations
- No formal senior testing yet — biggest gap
- English only, no other languages/dialects
- "Add someone you trust" is a placeholder
- Cost & latency at scale unvalidated

Grounded in: Lim & Tan (2003) · Ma (2023) · Zhai et al. (2025) · LaRubbio et al. (2025) · Roy & Nilizadeh (2024) · Singapore Police Force Annual Scam & Cybercrime Brief 2025